High-Resolution, Long-Distance Neural Circuit Manipulation and Imaging

Filip Tomaska, Che-Hang Yu, Xing Gao, Joseph Canzano, and Spencer LaVere Smith

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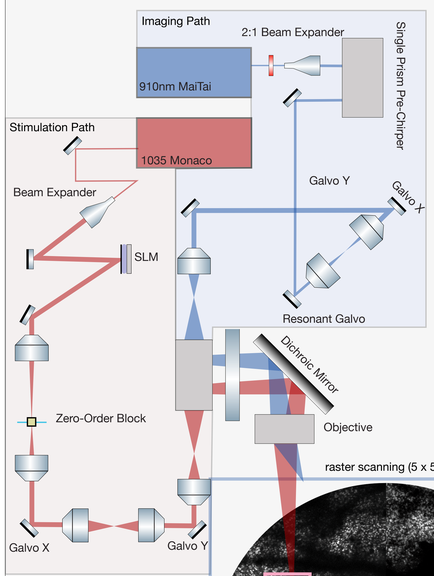
Introduction

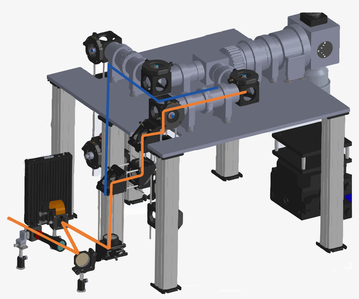
Multiphoton imaging and two-photon optogenetic stimulation provide flexible ways to measure and manipulate neuronal activity. To date, one of the main **challenges** for all-optical interrogation of neuronal circuits is the achievable **scale**. State-of-the-art spatial light modulators (**SLMs**) are limited to sub milimeter volume and are unable to generate holograms targeting distant parts of large FOVs (2 mm).
In this work, we present intrumentation that surpasses this constraint, enabling two-photon optomanipulation over a ***25 mm²*** FOV, with concurrent calcium imaging.

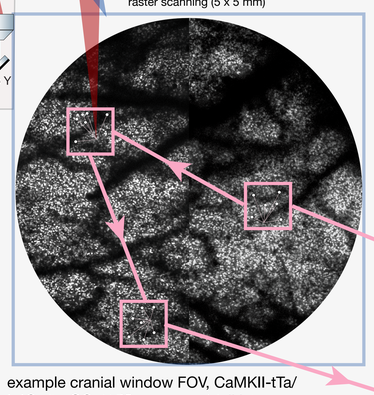
1. SLM-equipped Diesel2p is capable of imaging and optostimulation

In this configuration, Diesel2p offers **unconstrained** and **independent** access to the whole FOV for both imaging and optostimulation path. The volume available for holographic stimulation with the fast-refresh rate 1024x1024 SLM is **700 x 700 x 400 µm**. Both the imaging and stimulation ROI can be moved anywhere within the ***25 mm²*** FOV independently.

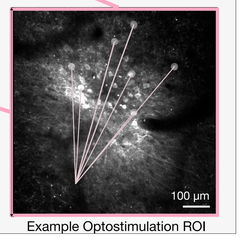
SLM-Diesel2p configuration schematic







example cranial window FOV, CaMKII-tTa/ tetO-riboGCaMP8m mouse, Niell lab



Example Optostimulation ROI

- **Imaging Laser**: 910 nm, 80 Hz (Mai-Tai, Newport)
- **Stimulation Laser**: 40 W 1035 nm low-repetition rate laser (Monaco, Coherent)
- 1 kHz 1024 x 1024 SLM with a clear aperture of 17.4 mm in plane conjugate to the imaging plane (Meadowlark)
- Relay system demagnifying the SLM by a factor of two to match the objective's pupil (32 mm)
- Diesel2p is an air objective with 8 mm of working distance allowing for non-horizontal sample imaging and stimulation.
- Maximal ROI dimensions: 1.5 mm x 5 mm (1024 X 4096) at 3.85 frames/s for one resonant arm.

References
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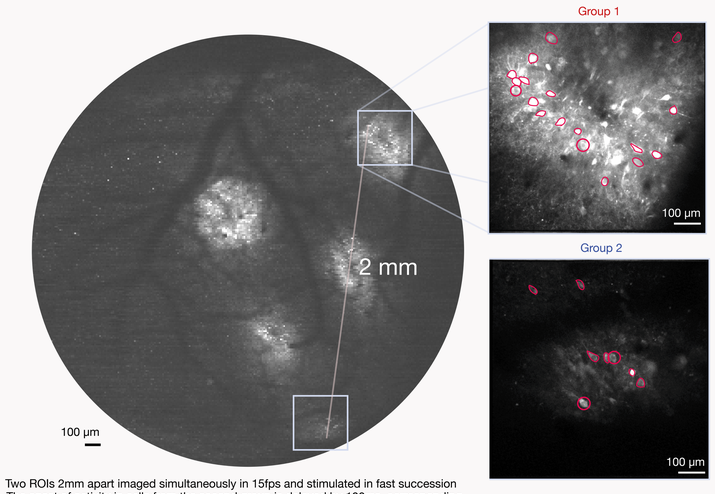
Summary

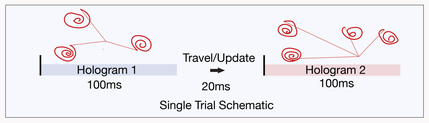
In this work, we adressed the problem of spatial scale mismatch between state of the art inter-areal calcium imaging and optostimulation. Our solution uses off-the shelf optics integrated into the existing design of Diesel2p without comrpromising its imaging capabilities. It is fully compatible with ScanImage for hologram generation and stimulation control. This configuration allows for unconstrained access to the whole 25mm² FOV for both imaging and photostimulation independently, without the need to move the sample, while matching the performance reported by other groups

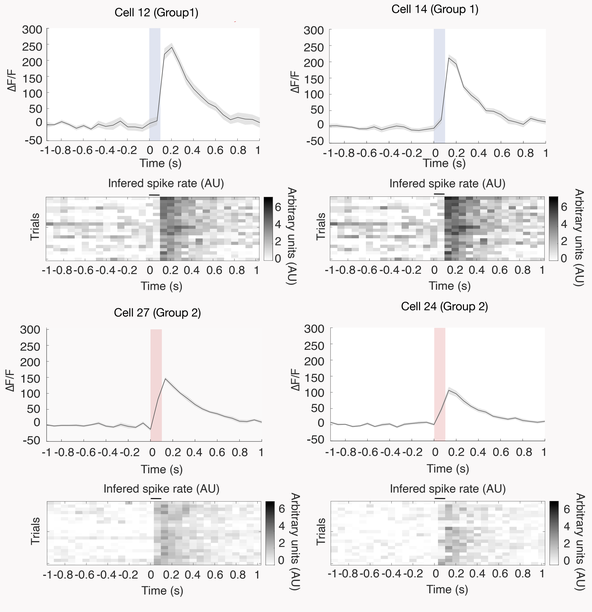
2. Fast and reliable long-distance stimulation

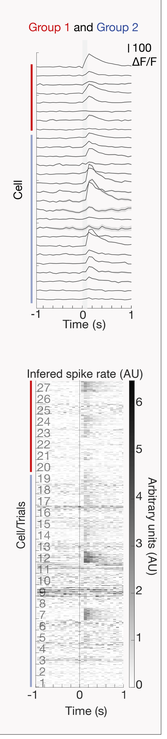
- Two cell populations (~15 cell holograms) located **~2 mm** apart can be stimulated in fast succession.
- Both travel time and phase mask update happens within a **~20 ms** window
- Most targeted cells are reliably responsive



Two ROIs 2mm apart imaged simultaneously in 15fps and stimulated in fast succession The onset of activity in cells from the second group is delayed by 100ms, corresponding to travel and refresh time < 67ms. Hardware trigger analysis measured the actual speed

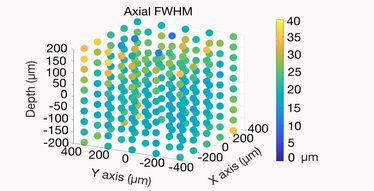


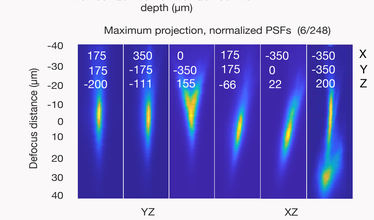


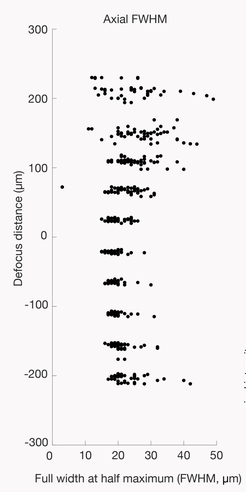


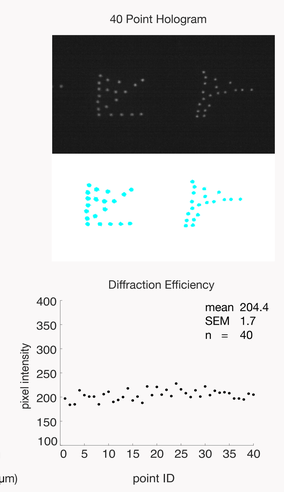
3. Characterization of holographic patterns using a substage camera

- **248 point cloud** cloud over a 700 x 700 x 400 µm volume was generated for characterization and calibration.
- PSFs of diffracted points were measured using a substage camera (81 slice zstack, 40 to -40 µm relative to the focal plane).
- Axial PSF FWHM **mean 21.9 µm (5.7 SD/0.26 SEM , n = 248**).
- A **40 point** diffuse hologram demonstrates uniformity of power distribution



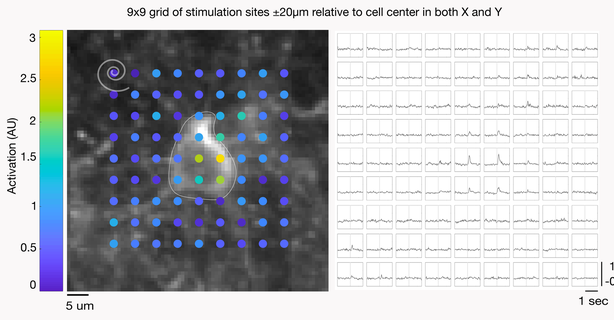


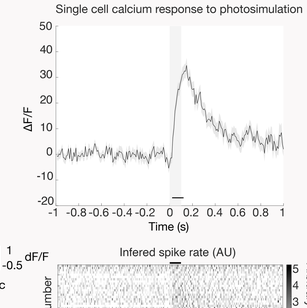


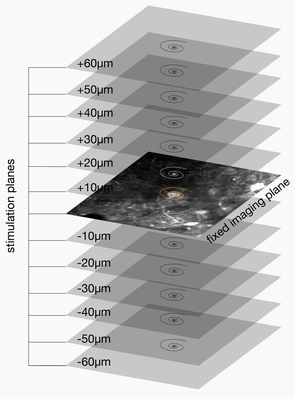


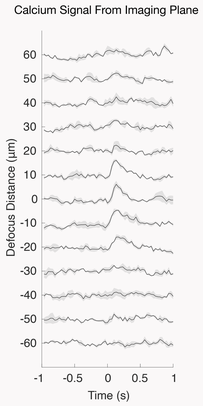
4. *In vivo* functional resolution of stimulation.

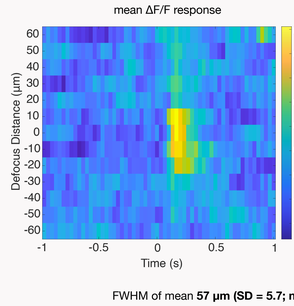
- Mouse model co-expressing virally delivered jGCaMP8m and ChRmine.
- A spiral scan stimulus (10 µm diameter, 10x10 ms) provokes a reliable calcium transient within tens of miliseconds (16%RF) .
- Same stimulus applied in a 9x9 grid (5µm spacing) reveals **~10 µm** functional lateral resolution
- A spiral scan applied in stimulation planes spanning from 60 to -60µm relative to the imaging plane measured functional axial resolution with FWHM of **mean 57 µm (SD=5.7, n=3)**











FWHM of mean **57 µm (SD = 5.7; n = 3)**